# On the nature of phonological cyclicity: Evidence from Nazarene Arabic*

Eyal Marco and Ezer Rasin
MIT and Tel-Aviv University

March 2026

**Abstract**

We present new evidence supporting the cycle as a grammatical mechanism in phonology. The evidence comes from the distribution of stress and vowel length in Nazarene Arabic, an understudied variety of Palestinian Arabic spoken in Nazareth. We show that the Nazarene Arabic pattern can be accounted for by cyclic versions of both rule-based phonology and Optimality Theory. The same pattern poses a challenge to Base-Derivative Correspondence – an alternative mechanism within Optimality Theory, according to which cyclic effects result from transderivational constraints that enforce similarity between related surface forms. We show that Base-Derivative Correspondence cannot account for the data, because the cyclic application of processes in Nazarene Arabic decreases rather than increases similarity between surface forms. Overall, this study highlights a divergent prediction of the cycle and Base-Derivative Correspondence, suggesting that phonological theory should include the former mechanism.

## 1 Introduction

The first generative theory of the phonology-morphology interface, proposed by Chomsky and Halle (1968) (following Chomsky, Halle, and Lukoff, 1956), stated that phonological operations apply to each word in a stepwise fashion, starting from the most deeply embedded constituent in the word's structure and proceeding outwards. This proposal, called *the cycle*, was motivated by cases where the phonology of morphologically complex words applies opaquely and systematically deviates from the regular phonology of the language (later named '*morphosyntactically induced opacity*' by Bermúdez-Otero 2018).

For example, mono-morphemic English words like *àbracadábra* or *wìnnipesáukee* typically have main stress on one of the last three syllables of the word and secondary stress on the first syllable. In contrast, morphologically complex words such as *Elìzabéthan* have surprising secondary stress on the second syllable of the word rather than the first. In a theory of the morphology-phonology interface that obeys the cycle, it is possible to predict the surprising location of secondary stress in *Elìzabéthan* from the position of main stress in *Elízabeth*, to which *Elìzabéthan* is related.

*Acknowledgments: to be added.

The logic is schematized in (1) and goes as follows. Phonology first applies to the stem *Elizabeth*, which starts its life as stressless, and assigns primary stress to the third syllable from the end, which is also the second syllable of the word. In a following step that includes the suffix *-an*, the phonology reassigns primary stress to *Elizabeth+an* while preserving – as secondary stress – the stress that was previously assigned to the second syllable of the stem *Elizabeth* (Chomsky and Halle 1968, Liberman and Prince 1977, Hayes 1982, Selkirk 1984, Bermúdez-Otero 2012). The cyclic derivation in (1) assigns primary stress to an abstract intermediate representation, |Elízabeth|, which undergoes further operations later in the derivation, when a larger morphological unit is considered.

(1) Derivation of *Elìzabéthan* in cyclic theories:
*Elizabeth* $\xrightarrow[\text{Phonology}]{}$ *Elízabeth* $\xrightarrow[\text{Morphology}]{}$ *Elízabeth-an* $\xrightarrow[\text{Phonology}]{}$ *Elìzabéthan*

While the original motivation for the cycle came from English, arguments for the cycle have since been made on the basis of numerous "cyclic patterns" from diverse languages, including Albanian (Trommer 2013), Finnish (Kiparsky 2003), Kashaya (Buckley 1994), Kimatuubmi (Odden 1996), Malayalam (Mohanan 1986), Maltese (Brame 1974), Moses Columbia Salish (Czaykowska-Higgins 1993), Palestinian Arabic (Brame 1974), Polish (Rubach 1984), San Mateo Huave (Noyer 2013), Sekani (Hargus 1988), Spanish (Brame 1974), Testsót'iné (Jaker and Kiparsky 2020), and many others.

Since the introduction of Optimality Theory (OT; Prince and Smolensky 1993), the cycle has been adopted by some serial extensions to OT, such as Stratal OT (Kiparsky, 2000, 2015; Bermúdez-Otero, 2011) and Cophonology Theory (Orgun, 1996; Inkelas and Zoll, 2007). In the classical OT model, however, a stepwise derivation is impossible because phonological processes are applied in a fully parallel fashion. Of course, in such a theory, abstract intermidiate representations are unavailable.

To simulate the cycle in Parallel OT, Benua (1997) relies on a typical property of cyclic patterns: the cyclic application of phonological processes often generates derivatives that resemble the surface forms of related words. For example, the surprising secondary stress in words like *Elìzabéthan* makes this word more phonologically similar to the word *Elízabeth* than if stress had fallen on the first syllable. Benua proposed that this similarity is not accidental, and is the result of a grammatical mechanism that actively enforces similarity between related words in the language. In particular, she proposes to extend faithfulness constraints from referring to just input-to-output correspondence (and base-reduplicant correspondence) to a new correspondence relation that holds between outputs, named Base-Derivative Correspondence (B-D Correspondence).[1]

Under B-D correspondence, the surprising secondary stress in words like *Elìzabéthan* overrides the default secondary stress pattern of English because of a grammatical correspondence requirement, according to which the surface form of this word should have similar stress to the surface form of the word *Elízabeth* which is related to it and serves as its *derivational base*. This is schematized in (2). B-D Correspondence typically defines the relevant notion of base and the way in which identity between the base and its morphological derivative is enforced.

[1]We will adopt the name B-D Correspondence for this kind of correspondence following Benua (1997). The proposal has also been named Transderivational Faithfulness, Output-to-Output Correspondence, and Paradigm Uniformity (see, e.g., Kenstowicz 1996; Steriade 1999; Kager 1999; Albright 2010).

(2) Derivation of *Elìzabéthan* using B-D correspondence:
*Elizabeth* $\xrightarrow[\text{Morphology}]{}$ *Elizabeth-an* $\xrightarrow[\text{Phonology}]{\text{Base: }\textit{Elízabeth}}$ *Elìzabéthan*

While both the cycle and B-D Correspondence achieve the correct result for the *Elìzabéthan* example, the competing theories make fundamentally different assumptions about the nature of phonological computation. According to the cycle, the similarity of the derivative form to its base is epiphenomenal – secondary stress is assigned to the second syllable of *Elìzabéthan* because primary stress first applies to the same syllable in the intermediate representation |Elízabeth|. In contrast, according to the B-D Correspondence account, the similarity between the forms is the very purpose of the phonological grammar. Under this view, secondary stress applies to the second syllable of *Elìzabéthan* so that this derivative would be more faithful to the surface form *Elízabeth*. The flowcharts in Figure 1 emphasize the architectural differences between the competing theories.[2]

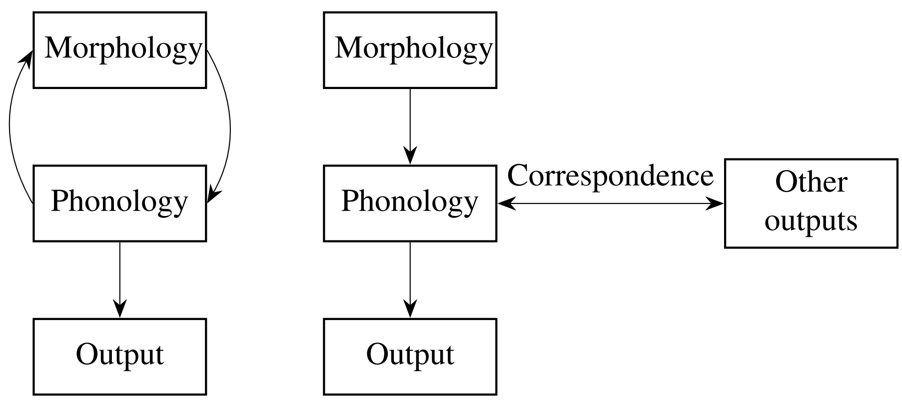


Figure 1: The cycle (left) vs. B-D Correspondence (right)

Given that the cyclic construction of morphologically complex words often leads to greater similarity between surface forms, it has been difficult to distinguish between the two theories in terms of their empirical predictions (for a few attempts, see Benua 1997; Kiparsky 2000, 2003, 2015; Bobaljik 2008; Steriade 2008; Bermúdez-Otero 2011, 2018; Trommer 2013; Gleim and Rasin 2024; Gleim 2025 among others).

In this paper, we analyze and discuss a pattern that can decide between the two alternatives. This pattern is cyclic in the sense that determining the phonology of some words seems to rely on the phonology of their parts in a way that is predictable by cyclic application. However, unlike most cyclic patterns, the cyclic application of processes in this case creates surface forms whose phonology is dissimilar to the phonology of related words. As we will see, this pattern can be accounted for by a serial theory obeying the cycle, but not by B-D correspondence, which is designed to simulate cyclicity in Parallel OT by enforcing similarity between surface forms.

[2]Our discussion ignores the difference between two models of cyclicity. The first is a truly modular approach, where morphology precedes phonology and phonology applies to the already-built morphological structure in a step-wise fashion, as proposed in SPE. The second is a model where morphology and phonology are intertwined, such that morphological operations can follow phonological operations, as proposed in Lexical Phonology & Morphology (Kiparsky, 1982b).

The pattern comes from the distribution of stress and vowel length in Nazarene Arabic (henceforth NZA), an understudied variety of Palestinian Arabic spoken in Nazareth. We will present the details of the full pattern in section 2, but for now just consider the small sample of data in (3).

(3) Examples illustrating the NZA pattern
   a. nsí:-na ‘we forgot’
   b. nsi-ná:-ha ‘we forgot her’
   c. nsi-na-há:-ʃ ‘we didn’t forget her’

The examples in (3) demonstrate a generalization regarding the distribution of stress and vowel length in NZA: a vowel is always long and stressed when it is immediately followed by the rightmost word-internal morpheme boundary. Otherwise, vowels are short and stressless before morpheme boundaries. This is true for all stems and suffixes in NZA without exception.

The NZA pattern can be accounted for in a cyclic theory by the interaction of three processes: vowel lengthening, stress, and vowel shortening. We will elaborate on this later, but for now, simplifying, assume the *Bracket Erasure* convention, according to which internal morpheme boundaries are erased at the end of every cycle (see Pesetsky 1979 and Kiparsky 1982a). In every cycle, lengthening applies to the vowel immediately preceding the morpheme boundary; then, stress is assigned; and then, shortening targets the pretonic vowel. As schematically illustrated in (4), the NZA pattern can be accounted for in this way and therefore supports the notions of the cycle and the Bracket Erasure convention. Here and below, $\varphi_1$, $\varphi_2$, and so on represent cycles and their corresponding morphological domains.

(4) Cyclic derivation of (3c) [nsinahá:ʃ] ‘we didn’t forget her’
   $[[[\text{nsi-na}]_{\varphi_1}\text{-ha}]_{\varphi_2}\text{-ʃ}]_{\varphi_3} \xrightarrow[\text{Cycle } \varphi_1]{} |\text{nsí:-na}| \xrightarrow[\text{Cycle } \varphi_2]{} |\text{nsiná:-ha}| \xrightarrow[\text{Cycle } \varphi_3]{} \text{nsinahá:-ʃ}$

As shown in (4), the cyclic application of the processes creates discrepancies between derivatives and their morphological bases. The long vowel in the output of the first cycle is shortened in the second cycle, while the word-final short vowel is lengthened. The same changes occur between the second cycle and the third cycle. As will be discussed in sections 3 and 4, this “ping pong” pattern poses a challenge to Parallel OT, because it requires referencing intermediate representations. This challenge cannot be resolved by B-D Correspondence because every step in the derivation decreases the similarity between the derivative and the output of the previous step. We therefore argue for the crucial properties involved in the derivation in (4), which are presented in (5).

(5) Architectural properties required to generate the NZA pattern
   a. A serial account of cyclicity.
   b. The ability to erase morphological boundaries between cycles.

This paper is structured as follows. In section 2, we first go through the relevant data and generalizations. Then, we provide a rule-based analysis of the pattern that manifests the properties in (5) and clarifies the argument to follow. In section 3, we present the challenge that the NZA pattern poses to Parallel OT, and in section 4, we show that B-D correspondence cannot help overcome this challenge. In section 5, we sketch a Stratal OT analysis of the pattern. Taken together, our theoretical sections show that serial accounts of cyclicity have no problem generating the NZA pattern. However, as we will see, the pattern cannot be generated without intermediate representations, even when B-D correspondence is employed. These sections teach us that cyclic effects are the result of a serial mechanism and are not a result of faithfulness between surface forms. In section 6, we present further data from NZA, showing that the pattern discussed in this paper is fully general and holds without exception across the language. Section 7 concludes.

# 2 Stress and vowel length in Nazarene Arabic

## 2.1 Basic generalizations

This section presents the core data on stress and vowel length in NZA, a variety of Palestinian Arabic spoken in Nazareth.[3] There has been much previous work on stress and vowel length in multiple colloquial dialects of Arabic, including varieties of Palestinian Arabic (Kenstowicz and Abdul-Karim, 1980; Abu-Salim, 1986; Younes, 1995; Watson, 2002; McCarthy, 2005, among others). The basic generalization regarding the distribution of stress is well established, and the basic facts about the distribution of vowel length have also been discussed.

The generalizations regarding stress and vowel length in NZA are not different from other varieties of Palestinian Arabic discussed in the literature. However, we chose to consult with NZA speakers in order to test the generality of the pattern beyond the examples discussed in the literature (below we present new data from dozens of morpheme combinations that have not been previously discussed). We were able to collect extensive data from three speakers of this particular variety and confirm that they all come from the same grammar (as opposed to data collected from different dialects with potentially inconsistent grammars; see de Lacy 2014 for discussion of this methodological point).

Turning to the generalizations, the stress pattern of NZA obeys the following algorithm (familiar from other dialects of Arabic; see Abdo, 1969; Brame, 1974; Watson, 2002):

(6) NZA stress pattern
  a. Stress the final syllable if superheavy (CVCC or CV:C).
  b. Otherwise, stress the penultimate syllable if heavy (CVC or CV:) or word-initial.
  c. Otherwise, stress the antepenultimate syllable.

The pattern is illustrated by the examples in (7). In (7a), the final syllable is a super-heavy CV:C syllable so it receives stress according to (6a). In (7b), the final syllable is a heavy CVC syllable, so (6a) does not hold. Instead, since the penultimate syllable is heavy, (6b) is met and penultimate stress is assigned. In (7c), the two final syllables are light CV syllables so neither (6a) nor (6b) are met and stress falls on the antepenultimate syllable by (6c). Beyond the examples in (7), the stress pattern is fully general and exceptionless.[4]

[3]The name NZA has not been commonly used in the phonological literature, which typically makes a coarse distinction between two varieties of Palestinian Arabic – "rural" and "urban". One clear difference between so-called "rural" and "urban" dialects is the pronunciation of the Modern Standard Arabic voiceless uvular stop [q], often pronounced as the glottal stop [ʔ] in "urban" and as the velar stop [g] in "rural". The variety spoken in Nazareth shows a mix of [ʔ] and [q] (the choice depends on sociological factors) and so does not fit into the common coarse classification. We therefore refer to it distinctly as "NZA".

[4]Though not always surface true, as stress can become obscured on the surface through vowel epenthesis. There has been extensive research on the interaction of stress and epenthesis in Arabic in the context of theory comparison (see, e.g., Broselow 1982, 2000; Kiparsky 2000). However, this interaction is independent of the theoretical discussion of the present paper, so we leave it aside.

(7) Examples illustrating the NZA stress pattern in (6)

a. mak.tab.téːn ‘two libraries’

b. mak.táb.tek ‘your.f.sg library’

c. mák.ta.be ‘library’

Vowel length can be unpredictable in stressed syllables, as illustrated by pairs of verbal templates that differ only with respect to the length of one of the vowels:

(8) NZA vowel length is contrastive in stressed syllables

a. CáCaC e.g., katab ‘he wrote’

b. Cá:CaC e.g., kaːtab ‘he corresponded’

In unstressed syllables, however, the distribution of vowel length is partially predictable. Several analyses have been proposed to account for this distribution in terms of a process of unstressed-vowel shortening. Abu-Salim (1986) proposed that unstressed vowels in open syllables are shortened before stressed vowels (e.g, [báːb] ‘*door*’ → [babéːn] ‘*two doors*’), but not before unstressed vowels (e.g, [ʕáːlam] ‘*world*’ → [ʕaːlaméːn] ‘*two worlds*’). He proposed a metrical analysis that enforces the shortening of pretonic vowels, and also correctly accounts for an opaque interaction between shortening and syncope (e.g, /ʃaːtir-iːn/ → [ʃaːtríːn] ‘*smart*.pl’). In a later study, Younes (1995) discussed northern rural dialects of Palestinian Arabic that are not completely compatible with Abu-Salim’s (1986) generalizations. Younes suggested that pretonic shortening applies to vowels in open syllables only when they precede long stressed vowels (but not when they precede short stressed vowels). The NZA facts seems to be compatible with his analysis.[5]

Consider the verbal paradigms of [ʃáːf] ‘*he saw*’ in NZA in (9). The table in (9a) shows the basic past inflection of the verb with all possible subject markers. The paradigms in (9b) show how three of the subjects markers can be combined with the object clitics of the language.

(9) Verbal paradigms of [ʃáːf] ‘*he saw*’[6,7]

a. Past subject inflections

| Subject | | |
|---|---|---|
| Infl | Affix | |
| 1.sg | -t | ʃúfet |
| 1.pl | -na | ʃúfna |
| 2.sg.f | -ti | ʃúfti |
| 2.sg.m | -t | ʃúfet |
| 2.pl | -tu | ʃúftu |
| 3.sg.f | -at | ʃáːfat |
| 3.sg.m | ∅ | ʃáːf |
| 3.pl | -u | ʃáːfu |

b. Past subject markers with object clitics

| Object | | Subject inflection | | |
|---|---|---|---|---|
| Infl | Affix | 3.sg.m | 3.sg.f | 3.pl |
| 1.sg | -ni | ʃáːfni | ʃaːfátni | ʃafúːni |
| 1.pl | -na | ʃáːfna | ʃaːfátna | ʃafúːna |
| 2.sg.f | -ek/-ki | ʃáːfek | ʃáːfatek | ʃafúːki |
| 2.sg.m | -ak/-k | ʃáːfak | ʃáːfatak | ʃafúːk |
| 2.pl | -ku | ʃáːfku | ʃaːfátku | ʃafúːku |
| 3.sg.f | -ha | ʃáːfha | ʃaːfátha | ʃafúːha |
| 3.sg.m | -o/-h | ʃáːfo | ʃáːfato | ʃafúː |
| 3.pl | -hen | ʃáːfhen | ʃaːfáthen | ʃafúːhen |

The 3.sg.m column in (9b) shows that the stem’s vowel remains long before an object clitic when that vowel is stressed (e.g., [ʃáːfni] ‘*he saw me*’). The 3.sg.f column shows that the stem’s vowel

[5]See Abu-Salim (1986) and Younes (1995) for further observations about vowel shortening in other environments that will not play a role in our paper.

[6]“Infl = Inflection. Our use of “Affix” does not make the traditional distinction between affixes and clitics.

[7]The 3.sg.m object marker [-h] can be deleted, as in /ʃaːf-u-h/ → [ʃafúː] ‘they saw him’.

also remains long before a stressed CVC syllable (e.g., [ʃaːfátni] '*she saw me*', as in Younes 1995, but not as in Abu-Salim 1986). The 3.pl column shows that the stem's vowel is shortened in open syllables before long stressed vowels (e.g., [ʃafúːni] '*they saw me*').

Subject and object markers may be followed by the negation morpheme /-ʃ/. Adding negation to the verbs in table (9b) reveals an environment where shortening overapplies to vowels that do not immediately precede a stressed syllable with another long vowel, as shown in (10). For example, while shortening applied right before a stressed long vowel in [ʃafúːni] 'they saw me' in (9b), the same vowel is also shortened in [ma ʃafuníːʃ] '*they did not see me*', even though the following syllable [fu] has a short, unstressed vowel. In section 2.2 we will analyze this overapplication as a cyclic effect, where shortening applies in an intermediate representation (ʃaːfúːni) before the negation morpheme is added. As can be seen in the 3.sg.f column in (10), the same shortening process does not apply to vowels that were not followed by long stress vowel in an intermediate representation (e.g, [ma ʃaːfatníːʃ] from [ʃaːfátni]). Therefore, we will propose that the overapplication of shortening involves application at an intermediate representation and cannot simply be explained as long-distance shortening.

(10) Past inflections of [ʃáːf] with object clitics and negation

| Object | | Subject inflection | | |
|---|---|---|---|---|
| Infl | Affix | 3.sg.m | 3.sg.f | 3.pl |
| 1.sg | -ni | ma ʃaːfníːʃ | ma ʃaːfatníːʃ | ma ʃafuníːʃ |
| 1.pl | -na | ma ʃaːfnáːʃ | ma ʃaːfatnáːʃ | ma ʃafunáːʃ |

Additional data show that the distribution of vowel length is also sensitive to the word boundary and to internal morpheme boundaries. Consider the verbal paradigms of [nísi] '*he forgot*' in (11). The paradigm in (11a) consists of the past subject inflections of the verb, and the paradigm in (11b) consists of all the combinations of the 1.pl subject affix with object clitics. Both paradigms show the verb with and without the negation affix. The verb [nísi] is conventionally analyzed with the root $\sqrt{n.s.j}$ whose glide consonant is only realized before vowels. Otherwise, the stem ends with a vowel.

(11) Verbal paradigms of [nísi] '*he forgot*'[8]

a. Past subject inflections

| Infl | Affix | Without Neg | With Neg |
|---|---|---|---|
| 1.sg | -t | nsíːt | ma nsíːteʃ |
| 1.pl | -na | nsíːna | ma nsináːʃ |
| 2.sg.f | -ti | nsíːti | ma nsitíːʃ |
| 2.sg.m | -t | nsíːt | ma nsíːteʃ |
| 2.pl | -tu | nsíːtu | ma nsitúːʃ |
| 3.sg.f | -at | nísjat | ma nisjáteʃ |
| 3.sg.m | ∅ | nísi | ma nisíːʃ |
| 3.pl | -u | nísju | ma nisjúːʃ |

b. Object clitics for 1.pl subjects

[8]The vowel [e] here is epenthesized before the negation suffix to break coda clusters.

| Infl | Affix | Without Neg | With Neg |
|---|---|---|---|
| 1.sg | -ni | nsináːni | ma nsinaníːʃ |
| 1.pl | -na | nsináːna | ma nsinanáːʃ |
| 2.sg.f | -ek/-ki | nsináːki | ma nsinakíːʃ |
| 2.sg.m | -ak/-k | nsináːk | ma nsinakáːʃ |
| 2.pl | -ku | nsináːku | ma nsinakúːʃ |
| 3.sg.f | -ha | nsináːha | ma nsinaːháːʃ |
| 3.sg.m | -h/-o | nsináː | ma nsinahúːʃ |
| 3.pl | -hen | nsináːhen | ma nsinahénneʃ |

Across the data in (11), stressless word-final vowels are short, illustrating the generalization in (12a) about NZA. In addition, as stated in (12b), a vowel immediately preceding the rightmost internal morpheme boundary is always long and stressed. All other vowels immediately preceding morphological boundaries are short and stresless.

(12) Generalizations regarding vowel length and morphological boundaries in NZA

a. Stressless word-final vowels are always short.

b. A vowel is always long and stressed when it is immediately followed by the rightmost word-internal morpheme boundary. Otherwise, vowels are short and stressless before morpheme boundaries.

Importantly, forms that do not obey the generalizations in (12) are ungrammatical. Hypothetical verbs with a short vowel before the rightmost morpheme boundary are ill-formed (e.g., 13a-13b), as well as hypothetical verbs with a long unstressed vowel before a morpheme boundary (e.g., 13c). A descriptively adequate phonological grammar must rule out these systematic gaps.

(13) Systematic gaps in NZA (cf. [nsinaháːʃ] '*we didn't forget her*')

a. *nsíː-na-h<u>a</u>-ʃ

b. *nsi-náː-h<u>a</u>-ʃ

c. *nsi-n<u>aː</u>-háː-ʃ

The generalizations in (12) can also be seen clearly in the imperfect paradigm (e.g, [btínsu] '*you.pl are forgetting*', [btinsúːha] '*you.pl are forgetting her*', and [btinsuháːʃ] '*you.pl are not forgetting her*'). In fact, the generalization holds without exception across dozens of different morpheme combinations, which to our knowledge are all possible NZA configurations in which the generality of the pattern can be tested. This is shown in section 6.

## 2.2 Analysis within rule-based phonology

In this section we present a rule-based analysis for the NZA pattern described above. We will provide an NZA grammar that can generate all of the data, as well as rule out the ungrammatical systematic gaps. We will show that such a grammar requires a serial account of cyclicity and the ability to erase morphological boundaries between cycles. Consider first the simple grammar in (14), which has two rules: STRESS, which assigns stress according to the algorithm in (6), and SHORTENING, which shortens vowels in open syllables before stressed long vowels.

(14) Grammar:

1. STRESS: assign stress according to (6)[9]
2. SHORTENING: V→ [-long] / ___ CVˊː

Consider first the derivation of [nsináːʃ] '*we didn't forget*'. As can be seen in the left column in (15), as long as all morpheme-final vowels are underlyingly long, such a grammar can generate the correct surface form. First, since the final syllable is super-heavy, stress is assigned to it. Only then, vowel shortening applies to the pretonic vowel and the correct output is derived.

(15) A rule-based analysis of NZA without LENGTHENING

| UR | /nsiː-naː-ʃ/ | /nsi-na-ʃ/ |
|---|---|---|
| STRESS | nsiː-náː-ʃ | nsí-na-ʃ |
| SHORTENING | nsi-náː-ʃ | - |
| SR | [nsináːʃ] | *[nsínaʃ] |

In contrast, the right column in (15) shows that if the morpheme-final vowels are underlyingly short, stress is assigned to the wrong vowel and an ungrammatical output is derived.

As will be discussed briefly in section 6, there is some evidence suggesting that the underlying representation (UR) of the stem is in fact /nisij/, and that the underlyingly first vowel and the final glide are deleted. For simplicity, we will only use the URs such as /nsi/ and /nsiː/ throughout the paper. The question whether one of these forms or /nisij/ is the actual UR does not interact with the pattern that we are interested in so we leave it aside.

As McCarthy (2005) already pointed out, and exemplified in (15), shortening by itself fails to rule out systematic gaps in Arabic, as we must take into account multiple URs. McCarthy's OT solution for this problem will be discussed later. Within rule-based phonology, a simple way to rule out this kind of systematic gaps is to add an early lengthening rule, given in (16), which lengthens any vowel followed by a morpheme boundary.

(16) LENGTHENING: V → [+long] / ___+

The derivations in (17) show that the three rules correctly derive [nsináːʃ] regardless of the underlying length specification of morpheme-final vowels (which alternate between long and short on the surface).

(17) Derivation of [nsináːʃ] from URs with different length specifications

| UR | /nsiː-naː-ʃ/ | /nsi-naː-ʃ/ | /nsiː-na-ʃ/ | /nsi-na-ʃ/ |
|---|---|---|---|---|
| LENGTHENING | - | nsiː-naː-ʃ | nsiː-naː-ʃ | nsiː-naː-ʃ |
| STRESS | nsiː-náː-ʃ | nsiː-náː-ʃ | nsiː-náː-ʃ | nsiː-náː-ʃ |
| SHORTENING | nsi-náː-ʃ | nsi-náː-ʃ | nsi-náː-ʃ | nsi-náː-ʃ |
| SR | [nsináːʃ] | [nsináːʃ] | [nsináːʃ] | [nsináːʃ] |

There are two further things to account for: the length of word-final vowels, and the overapplication of shortening in some environments. First, as shortening applies only to pretonic vowels, the grammar does not rule out words with a stressless long final vowel (e.g., *[nsináːhaː]), which contradict the generalization in (12a). Such surface forms can be ruled out straightforwardly by adding a rule that shortens word-final stressless vowels:

[9]A formal stress assignment rule can be stated as follows: V → [+stress] /___$C_0((VC)VC_0^1)\#$ (cf. Brame 1974).

(18) FINAL SHORTENING: $\underset{\text{[-stress]}}{\text{V}} \rightarrow$ [-long] / __#

This point will not play a role in our argument, but is discussed here to present a more complete analysis.

The second thing to account for is the opaque shortening in forms such as [nsinaháːʃ] *'we didn't forget her'*. In such configurations, the antepenultimate vowel is shortened even though there is no conditioning environment for shortening on the surface – the vowel does not immediately precede a long stressed vowel. As was also shown by Watson (2002) for a different Arabic variety, this can be accounted for by applying the rules cyclically. The derivations in (19) illustrate this.

(19) Cyclic derivations of [nsinaháːʃ] *'we didn't forget her'* and [ʃaːfatníːʃ] *'she didn't see me'*

| *Input Structure* | [[[nsi-na]$_{\varphi_1}$-ha]$_{\varphi_2}$-ʃ]$_{\varphi_3}$ | [[[ʃaːf-at]$_{\varphi_1}$-ni]$_{\varphi_2}$-ʃ]$_{\varphi_3}$ |
|---|---|---|
| | Cycle $\varphi_1$ | |
| *Input* | /nsi-na/ | /ʃaːf-at/ |
| LENGTHENING | nsiː-na | - |
| STRESS | nsíː-na | ʃáːf-at |
| SHORTENING | - | - |
| | Cycle $\varphi_2$ | |
| *Input* | /nsíːna-ha/ | /ʃáːfat-ni/ |
| LENGTHENING | nsíːnaː-ha | - |
| STRESS | nsiːnáː-ha | ʃaːfát-ni |
| SHORTENING | nsináː-ha | - |
| | Cycle $\varphi_3$ | |
| *Input* | /nsináːha-ʃ/ | /ʃaːfátni-ʃ/ |
| LENGTHENING | nsináːhaː-ʃ | ʃaːfátniː-ʃ |
| STRESS | nsinaːháː-ʃ | ʃaːfatníː-ʃ |
| SHORTENING | nsinaháː-ʃ | - |
| *Final Output* | nsinaháːʃ | ʃaːfatníːʃ |

As can be seen in the derivation of [nsinaháːʃ] in (19), shortening of the antepenultimate vowel is triggered in the second cycle because in the intermediate representation |nsiːnáː-ha|, the penultimate vowel is long and stressed. In contrast, in the derivation of [ʃaːfatníːʃ] the antepenultimate vowel never appears before a long and stressed vowel throughout the derivation and therefore remains long on the surface. For simplicity, (19) does not show derivations for all conceivable underlying length specifications of morpheme-final vowels. However, the correct output would be derived given all the URs in (17).

The derivations in (19) rely on and therefore support Pesetsky's (1979) notion of the Bracket Erasure convention, given in (20). The cyclic application of the three rules derives the correct outputs because, as (20) determines, only the most external internal morpheme boundary is available in every cycle and, therefore, constitutes the only trigger for vowel lengthening before morpheme boundaries. If nested morpheme boundaries were not erased, the antepenultimate vowel in [nsi-na-háː-ʃ], which is crucially shortened in the second cycle, would have been lengthened again in the third cycle, generating the incorrect form *[nsiː-na-háː-ʃ].

(20) The *Bracket Erasure* convention (cf. Pesetsky 1979)
Given the nested constituent $[\cdots[\cdots]_{\varphi_{n-1}}\cdots]_{\varphi_n}$,
the last operation on cycle $\varphi_n$ is to erase the brackets of $\varphi_{n-1}$.

The Bracket Erasure convention is challenged by some phonological patterns that have been argued to require the persistence of internal morpheme boundaries throughout the derivation. In response, the definition in (20) has been revised from a universal convention to a language-specific condition, and weaker versions of (20) have been proposed as alternatives (see, e.g., Hargus 1988; Orgun and Inkelas 2002; Jaker and Kiparsky 2020). Our study is not informative for the discussion about the cross-linguistic status of Bracket Erasure. This paper only provides evidence that phonological theory should at least allow for erasing internal morpheme boundaries at the end of every cycle.[10]

To conclude the insights provided by the rule-based analysis of the NZA pattern, consider again the outputs of every cycle in the derivation of [nsinaháːʃ] *'we didn't forget her'* in (19), which were presented in (4) and repeated here in (21).

(21) $[[[\text{nsi-na}]_{\varphi_1}\text{-ha}]_{\varphi_2}\text{-ʃ}]_{\varphi_3} \xrightarrow[\text{Cycle } \varphi_1]{} |\text{nsíː-na}| \xrightarrow[\text{Cycle } \varphi_2]{} |\text{nsináː-ha}| \xrightarrow[\text{Cycle } \varphi_3]{} \text{nsinaháː-ʃ}$

As mentioned in the introduction and shown in (21), the cyclic application of the processes creates discrepancies between derivatives and their bases. The long vowel in the output of the first cycle (ns**<u>íː</u>**-na) is shortened in the second cycle (ns<u>i</u>náː-ha), while the word-final short vowel in the output of the first cycle (nsíː-n<u>a</u>) is lengthened in the second cycle (nsin**<u>áː</u>**-ha). The same changes occur between the second cycle and the third cycle: the long vowel is shortened (nsin**<u>áː</u>**-ha → nsin<u>a</u>háː-ʃ), while the word-final short vowel is lengthened (nsináː-h<u>a</u> → nsinah**<u>áː</u>**-ʃ). As we will see next, this "ping-pong" pattern poses a challenge to parallel OT which, unlike other cyclic patterns, cannot be resolved by B-D correspondence.

# 3 The challenge for Parallel OT

## 3.1 Standard stress constraints

The main takeaway from the previous section is the following. In a cyclic architecture, and as long as internal morpheme boundaries are erased at the end of every cycle, the NZA pattern can be accounted for straightforwardly. This is because, according to the rule-based analysis we have just seen, vowel length is sensitive both to morphological boundaries and to stress in intermediate representations. In each cycle, the vowel preceding the sole internal morpheme boundary present in that cycle is lengthened; then, stress is assigned according to the NZA stress algorithm; and then, shortening applies to the pretonic vowel if the stressed vowel is long. In section 5, we will see that Stratal OT, a cyclic version of OT, has no problem accounting for the NZA pattern as well. Similarly to the rule-based analysis, the Stratal OT analysis also relies on intermediate representations and can erase internal morpheme boundaries between cycles.

[10]Our study, however, supports Pesetsky's (1979) revision of the original proposal from the SPE (Chomsky and Halle, 1968), according to which internal morpheme boundaries are erased *at the beginning* of every cycle. As shown in (19), the rightmost internal morpheme boundary must be present during the cycle to trigger vowel lengthening. As Pesetsky proposes, the boundaries should be erased at the end of the cycle to avoid triggering the same process again.

The classical version of OT, however, computes stress and vowel length in parallel, without intermediate representations, and thus has a problem assigning stress to the right position. Usually, B-D correspondence is used to account for cyclicity without serialism (as in, for example, McCarthy's 2005 analysis, to be reexamined in section 4.2). We will see in section 4, however, that B-D correspondence will not solve the challenge posed by the NZA pattern. The reason is that in this particular case, the cyclic application of processes does not increase the similarity between derivatives and their morphological bases. The conclusion will be that phonological theory must include a serial account for cyclicity, rather than trying to imitate cyclicity using B-D correspondence.

Let us see the initial challenge for Parallel OT without B-D correspondence coming from stress and vowel length. The challenge is to choose the output in (22a) over the incorrect but prosodically well-formed alternatives in (22b) and (22c) without causing trouble elsewhere. To show the problem, we will consider various choices for stress constraints and their ranking.

(22) Correct output vs. problematic alternatives

a. [nsi-na-háː-ʃ] (correct)

b. *[nsi-náː-ha-ʃ] (incorrect)

c. *[nsíː-na-ha-ʃ] (incorrect)

Consider first a standard choice of stress constraints from Prince (1990) and Prince and Smolensky (1993) and a ranking of these constraints, given in (23), that could generate stress as in (6) after certain additions and refinements that will be left aside to keep the discussion simple.

(23) $\text{WSP}_{\text{Super-heavy}} \gg$ NONFINALITY $\gg \text{WSP}_{\text{Heavy}} \gg$ EXTNONFINALITY $\gg$ ...

The constraints in (23) are EXTNONFINALITY, which penalizes stress on either of the final two syllables, $\text{WSP}_{\text{Heavy}}$, which requires heavy syllables (CVC, CV:) to be stressed, NONFINALITY, which penalizes stress on the final syllable, and $\text{WSP}_{\text{Super-heavy}}$, which requires super heavy syllables (CVCC, CV:C) to be stressed. The ranking (again, after some additions and refinements) would ensure that stress on the ante-penultimate syllable is the default, but this default can be overridden if the penultimate syllable is heavy (in which case stress will be penultimate) or if the final syllable is superheavy (in which case stress will be final).

Because NONFINALITY and EXTNONFINALITY are violated by ultimate and penultimate stress, this ranking favors antepenultimate stress, which means that among the three output candidates in (22), it favors an incorrect candidate, as schematized in (24). Here we use URs with morpheme-final long vowels to illustrate the problem. Of course, given the principle of Richness of the Base (Prince and Smolensky, 1993; Smolensky, 1996), the correct output needs to be selected given any underlying specification of length. We will return to this point later on.[11]

(24)

| /nsiː-naː-haː-ʃ/ | $\text{WSP}_{\text{Super-heavy}}$ | NONFINALITY | $\text{WSP}_{\text{Heavy}}$ | EXNONFIN |
|---|---|---|---|---|
| a. ☹ nsi-na-háː-ʃ | | *! | | * |
| b. nsi-náː-ha-ʃ | | | | *! |
| c. ☞ nsíː-na-ha-ʃ | | | | |

[11] As mentioned before, we do not use the UR /nisij/ for the stem to simplify the analysis. This choice does not bear on the discussion here, and only allows us to ignore the constraints responsible for the deletions of the first vowel and final glide in /nisij/.

For a Parallel OT analysis to work, some other constraint (or constraints) $\mathbb{C}$ that favors the correct candidate (24a) over candidates (24b) and (24c) must outrank NONFINALITY. With such a constraint in place, candidate (24a) will win, as illustrated schematically in the following tableau:

(25)

| | | /nsiː-naː-haː-ʃ/ | $\mathbb{C}$ | $\text{WSP}_{\text{Super-heavy}}$ | NONFINALITY | ... |
|---|---|---|---|---|---|---|
| a. | ☞ | nsi-na-háː-ʃ | | | * | ... |
| b. | | nsi-náː-ha-ʃ | *! | | | ... |
| c. | | nsíː-na-ha-ʃ | *! | | | ... |

Which constraint can $\mathbb{C}$ be? We will go over several options that do not work before mentioning an option that works technically and discussing its limitations. First, $\mathbb{C}$ cannot be simple markedness constraints that are independently motivated by the distribution of length in NZA. Consider the constraints in (26):

(26)
- $^*\breve{V}+$ - assign * for every short vowel that is immediately followed by a morpheme boundary.
- $^*V{:}C\acute{V}{:}$ - assign * for every long vowel in an open syllable that precedes a stressed long vowel.

The first constraint in (26) can motivate the lengthening of vowels before morpheme boundaries, and the second can motivate pretonic shortening. These independently motivated constraints, however, cannot account for the choice of the locus of stress and lengthening. As can be seen in (27), the candidates from (24) are equally harmonic with respect to these markedness constraints. Because Bracket Erasure relies on cycles, which are not available in Parallel OT, all internal morpheme boundaries are included in all candidates. Therefore, each one of the candidates in (27) violates $^*\breve{V}+$ twice.

(27)

| | /nsiː-naː-haː-ʃ/ | $^*V{:}C\acute{V}{:}$ | $^*\breve{V}+$ | ... |
|---|---|---|---|---|
| a. | nsi-na-háː-ʃ | | ** | ... |
| b. | nsi-náː-ha-ʃ | | ** | ... |
| c. | nsíː-na-ha-ʃ | | ** | ... |

$\mathbb{C}$ cannot be a familiar input-output faithfulness constraint either. As mentioned above, the correct pattern should be generated given URs with any specification of vowel length. Therefore, any attempt to favor [nsi-na-háː-ʃ] over *[nsi-náː-ha-ʃ] and *[nsíː-na-ha-ʃ] that refers to the underlying length in /nsiː-naː-haː-ʃ/ will not work for a different UR where the same vowels are short. For the same reason, faithfulness to underlying stress will be of no help.

Consider now the possibility that $\mathbb{C}$ is a more complex markedness constraint that is not independently motivated in NZA. To understand the degree of specificity required from this constraint, consider the words in (28), which have an underlying long vowel.

(28) a. miʃwáːr-ak 'your.m.sg walk' (cf. miʃwar-éːn 'two walks')
b. ʔistaʃáːr-ek 'he consulted you.f.sg' (cf. ʔistaʃar-úːni 'they consulted me')
c. ʕáːlam-ak 'your.m.sg world' (cf. ʕaːlam-éːn 'two worlds')

Given the prosodic well-formedness of these words in NZA, $\mathbb{C}$ would have to penalize the candidate *[nsi-náː-ha-ʃ] without also penalizing (28a)-(28b). Similarly, it has to penalize *[nsíː-na-ha-ʃ] without penalizing (28c). This is because, the choice of the candidate [nsi-na-háː-ʃ] over *[nsi-náː-ha-ʃ] and *[nsíː-na-ha-ʃ] under a ranking with such a high-ranking markedness constraint $\mathbb{C}$ would teach us that $\mathbb{C}$ can be satisfied by stressing the final syllable and at the same time lengthening a final underlyingly short vowel. If $\mathbb{C}$ also penalized (28a)-(28c), the fact that $\mathbb{C}$ can be repaired by stressing the final syllable and lengthening the final vowel would mean, for example, that the ungrammatical *[miʃwar-áːk] should win over [miʃwáːr-ak]).

To distinguish *[nsi-na-háː-ʃ] from (28a)-(28c), $\mathbb{C}$ could be a markedness constraint that refers to morpheme boundaries, which are the main surface difference between *[nsi-náː-ha-ʃ]/*[nsíː-na-ha-ʃ] and (28a)-(28c). A markedness constraint that penalizes *[nsi-náː-ha-ʃ] and *[nsíː-na-ha-ʃ] but not [nsi-na-háː-ʃ] or (28a)-(28c) and refers to morpheme boundaries is the one in (29):

(29) $^{*}\breve{V} + \underbrace{\{C,V\}^{*}}_{No\ +}\#$: assign * for a short vowel at the end of the penultimate morpheme of the word (whatever the morphology of the word is)

With this constraint in place, the correct candidate is the winner:

(30)

| | | /nsiː-naː-haː-ʃ/ | $^{*}\breve{V} + \{C,V\}^{*}\#$ | WSP$_{\text{Super-heavy}}$ | NONFINALITY | ... |
|---|---|---|---|---|---|---|
| a. | ☞ | nsi-na-háː-ʃ | | | * | ... |
| b. | | nsi-náː-ha-ʃ | *! | | | ... |
| c. | | nsíː-na-ha-ʃ | *! | | | ... |

The major problem with (29) is not its complexity or its lack of independent motivation in NZA, but rather that it makes pathological typological predictions that theories without it can avoid. Examples of processes that (29) can generate (by appropriately ranking it with respect to faithfulness constraints) include the following:

(31) Pathological typological predictions made by (29)

a. Delete the vowel in the penultimate morpheme of the word if that vowel is short.

b. Insert a consonant after the vowel in the penultimate morpheme of the word if that vowel is short.

Such processes violate a broader generalization regarding attested phonological processes that has been implicitly assumed to hold by phonologists for decades. By including both $+$ and $\#$ in its description, (29) causes a direct violation of the following phonological universal:

(32) **Universal** (entailed by the proposals of Chomsky and Halle 1968, Pesetsky 1979, Kiparsky 1982b, Orgun and Inkelas 2002, and Bermúdez-Otero 2011):
A phonological process cannot refer to both a word boundary and an internal morpheme boundary.[12]

If the universal in (32) is correct, constraints like (29) should be excluded from the space of possible constraints in OT.[13]

[12] Processes that refer to both a word boundary and an internal morpheme boundary were originaly ruled out by the Bracket Erasure convention. As discussed in section 2.2, the universality of Bracket Erasure is under debate. However, (32) still follows by a weaker version of the convention, which is defended by Orgun and Inkelas (2002).

[13] Note that the problem with (29) is independent of whether OT constraints are innate or acquired. As long as (29) can be stated in the grammar and be ranked above the relevant faithfulness constraints, the patterns in (31) would be generable.

We can conclude that an analysis of the NZA pattern in terms of (29) results in unwelcome typological predictions in different degrees of severity. The pathologies in (31) already seem like bad enough predictions, and until counterexamples to the universal in (32) are found, the problem for (29) is even more general. The rule-based analysis from section 2.2 as well as the Stratal OT analysis that will be presented in section 5 do not suffer from the same problem. By positing intermediate representations, those analyses account for the distribution of stress in a simple way and do not have to resort to rules or constraints like (29).

## 3.2 Alternative stress constraints

The discussion so far has assumed the stress constraints in (23) that make non-final stress the default. A default of non-final stress is not an accident of that set of constraints, but rather a general consequence of how complementarity is typically analyzed within OT (see Baković 2013 for much relevant discussion): the occurrence of an element in a specific environment (e.g., voiced obstruents only intervocalically) is enforced using a markedness constraint penalizing the absence of that element in that environment (e.g., *VTV in the case of intervocalic voicing). The reverse – penalizing the element in the elsewhere set of environments – would often require a complex list of constraints with potentially problematic typological consequences (e.g., constraints penalizing voiced obstruents word-initially in the case of intervocalic voicing). The stress pattern of NZA is no different: the effect of weight on stress (final stress only on super-heavy syllables, otherwise penultimate stress only on heavy syllables, otherwise antepenultimate stress) motivates a constraint hierarchy that makes antepenultimate stress (the elsewhere case) the default, and that uses constraints like $\text{WSP}_{\text{Heavy}}$ and $\text{WSP}_{\text{Super-heavy}}$ to push stress rightwards precisely when heavier syllables follow.

Suppose, however, that some reasonable constraints could be devised that account for the complementarity in NZA stress by making rightmost stress the default and by deflecting stress leftwards away from syllables that are too light. One conceivable set of ranked constraints that can generate the NZA stress pattern and has this property is the following: $\text{NONFINALITY}_{\text{Light}}$ (penalize a final light stressed syllable), $\text{NONFINALITY}_{\text{Heavy}}$ (penalize a final heavy stressed syllable), $\text{EXTNONFINALITY}_{\text{Light}}$ (penalize a penultimate light stressed syllable), and ALIGNR (stress is aligned to the right), where ALIGNR is dominated by the three other constraints. We will now show that the NZA pattern poses a problem for Parallel OT even under such constraints. By doing that, we will have shown that the problem for Parallel OT discussed in the previous sections is more general and did not arise because of the choice of the stress constraints in (23).

Consider again the tableau for /nsiː-naː-haː-ʃ/ from (24), repeated in (33) with the alternative stress-markedness constraints just mentioned, which this time make final stress the default. Since final stress is the default, the incorrect candidates (33b) and (33c) receive fatal violations and the correct candidate (33a) is now chosen.

(33)

| /nsiː-naː-haː-ʃ/ | $\text{NONFIN}_{\text{Light}}$ | $\text{NONFIN}_{\text{Heavy}}$ | $\text{EXTNONFIN}_{\text{Light}}$ | ALIGNR |
|---|---|---|---|---|
| a. ☞ nsi-na-háː-ʃ | | | | |
| b. nsi-náː-ha-ʃ | | | | *! |
| c. nsíː-na-ha-ʃ | | | | *!* |

This analysis, while correctly generating [nsi-na-háː-ʃ], creates a new problem elsewhere. Given

Richness of the Base, the correct candidate (33a) must be generated even when the underlying form has a penultimate vowel followed by a short vowel, i.e., /nsi-naː-ha-ʃ/. For candidate (33a) to win over candidate (33b) as an output of /nsi-naː-ha-ʃ/, the pressure for final stress must be strong enough to enforce final stress even at the expense of lengthening and shortening underlying vowels. This, in turn, makes the incorrect prediction that words like those mentioned above in (28), such as [miʃwáːr-ak] (from underlying /miʃwaːr-ak/, which have the same prosodic profile as candidate (33b), should surface with final stress and a final long vowel, contrary to fact:

(34) a.

| | | /nsi-naː-ha-ʃ/ | ... | ALIGNR | IDENT(long) | ... |
|---|---|---|---|---|---|---|
| a. | ☞ | nsi-na-háː-ʃ | | | ** | ... |
| b. | | nsi-náː-ha-ʃ | | *! | | ... |

b.

| | | /miʃwaːr-ak/ | ... | ALIGNR | IDENT(long) | ... |
|---|---|---|---|---|---|---|
| a. | ☹ | miʃwáːr-ak | | *! | | ... |
| b. | ☞ | miʃwar-áːk | | | ** | ... |

To save Parallel OT, then, the challenge is to find a grammar that generates the correct [nsi-na-háː-ʃ] without also over-generating ungrammatical alternatives to words like [miʃwáːr-ak].

The conclusion for Parallel OT is that the theory is unable to account for the NZA pattern. The failure is due to the full parallelism of the theory and its rejection of the intermediate representations that cyclic patterns require. As will be shown immediately, [nsi-na-háː-ʃ] and similar forms cannot be selected using B-D correspondence constraints either.

# 4 B-D Correspondence cannot solve the challenge

## 4.1 The challenge for B-D correspondence

As discussed in the introduction, B-D correspondence had empirical success in accounting for cyclic patterns in Parallel OT. This success relies on the fact that the cyclic application of processes often increases the similarity between related words. However, this is not the case for the 'ping pong' pattern we examine here. We therefore argue that the NZA pattern constitutes evidence that similarity between derivatives and their bases is an epiphenomenon and is not the purpose of phonological computation. In what follows we will see that B-D correspondence indeed fails to replace cyclicity when it comes to the NZA pattern.

Consider first the tableau in (35). The tableau shows the derivation of the input /nsiː-naː-haː-ʃ/, given the stress related constraints used in section 3.1, and a higher ranked B-D faithfulness constraint: BD-IDENT. The constraint penalizes discrepancies in vowel length between corresponding segments in a derivative form and its base.

(35)

| | | /nsiː-naː-haː-ʃ/ *Base*: nsináːha | BD-IDENT | WSP Super-heavy | NON FINALITY | WSP heavy | EXNON FINALITY |
|---|---|---|---|---|---|---|---|
| a. | ☹ | nsi-na-háː-ʃ | *!* | | * | | |
| b. | ☞ | nsi-náː-ha-ʃ | | | | | * |
| c. | | nsíː-na-ha-ʃ | *!* | | | | |

As can be seen in (35), a B-D faithfulness constraint cannot solve the challenge because candidate (35b), *[nsi-náː-ha-ʃ], is strictly more similar to the base ([nsináːha]) than to the correct output form in (35a), [nsi-na-háː-ʃ].

As was emphasized in section 2.2, the reason for the failure of B-D correspondence in accounting for this pattern is that the cyclic application of lengthening, stress, and shortening in NZA leads to changes in vowel length between derivatives and their bases. Therefore, cyclic derivatives in NZA cannot be a product of satisfying faithfulness constraints. This is the property that distinguishes between the NZA pattern and other cyclic patterns that motivated the introduction of B-D correspondence to phonological theory (see, Benua 1997; Kager 1999; Steriade 1999 among others).

This is not a challenge that can be resolved by different B-D constraints than the one used in (35). Stanton and Steriade (2014) propose that faithfulness constraints can also require uniformity between derivatives and non-local bases. However, there is no benefit in turning to the remote base, [nsíːna], because it most resembles the wrong candidate in (35c). Anti-faithfulness B-D constraints (Alderete 2001) cannot help either. These constraints are satisfied by violations of the faithfulness constraints they negate. As can be seen in (36), an Anti-faithfulness constraint (¬BD-IDENT), which compels derivatives to be unfaithful to their bases, can rule out candidate (36b). However, such a constraint cannot distinguish between (36a) and (36c) because both candidates incur the same faithfulness violations relative to the base ([nsináːha]). The correct output candidate in (36a) still loses because of the stress related constraint that prefers antepenultimate stress other things being equal.

(36)

| | | /nsiː-naː-haː-ʃ/ *Base*: nsináːha | ¬BD-IDENT | WSP Super-heavy | NON FINALITY | WSP heavy | EXNON FINALITY |
|---|---|---|---|---|---|---|---|
| a. | ☹ | nsi-na-háː-ʃ | | | *! | | |
| b. | | nsi-náː-ha-ʃ | *! | | | | * |
| c. | ☞ | nsíː-na-ha-ʃ | | | | | |

Expanding the theory to allow anti-faithfulness to non-local bases (i.e., adopting a complex combination of the mechanisms proposed by Alderete 2001 and Stanton and Steriade 2014) is also futile. Candidate (35b) is the least faithful candidate relative to the remote base ([nsíːna]) and therefore will be favored by an anti-faithfulness constraints that is satisfied by faithfulness violations relative to the remote base. The failure of anti-faithfulness further shows that a decreased similarity between surface forms, as in the NZA pattern, is also an epiphenomenon and not a direct grammatical goal.

The conclusion is that even with B-D correspondence, the extension to OT designed to account for cyclic patterns without serialism, Parallel OT cannot generate the NZA pattern. In the next section we will see that a version of OT assuming a serial account of cyclicity and Bracket Erasure does not face the same challenge. Before we see it, we will discuss the shortcomings of a previous account for a similar pattern, proposed by McCarthy (2005).

## 4.2 A reexamination of McCarthy's (2005) analysis

McCarthy (2005) offers an analysis of the distribution of stem-final vowels in Cairene Arabic, which is similar to the NZA pattern discussed here. McCarthy incorporated B-D correspondence as part of his analysis, and it is therefore important that we examine his proposal before moving forward. We will show that his analysis succeeds on pre-boundary vowels in words that have one suffix, but breaks down when there is more than one suffix.

McCarthy points out that shortening by itself is not sufficient to rule out systematic gaps in Cairene Arabic, akin to the NZA gaps in (13). Therefore, he proposes that an underlying long vowel is shortened word-finally, but also that that an underlying short vowel is deleted in the same position. McCarthy uses B-D correspondence to make sure that a stem-final short vowel will be deleted before a suffix as well, while a stem-final long vowel will surface before a suffix. His analysis uses the constraints in (37), as well as the stress constraints NONFINALITY and WSP that were presented above.

(37) McCarthy's (2005) constraints

- MAX-$\mu$ - assign * for every mora in the input that has no correspondent in the output.
- MAX-Vː - assign * for every long vowel in the input that has no correspondent in the output.
- FIN-C - assign * for every word final vowel.
- BD-DEP-V - assign * for every vowel in the output that has no correspondent in the base's output.

To see how the analysis works, consider first the unsuffixed forms in (38) and (39).

(38) McCarthy's (2005) derivation of a stem-final short vowel

| | | /takaba/ | NONFIN | WSP | MAX-Vː | BD-DEP-V | FIN-C | MAX-$\mu$ |
|---|---|---|---|---|---|---|---|---|
| a. | ☞ | tákab | | | | | | * |
| b. | | tákaba | | | | | *! | |

(39) McCarthy's (2005) derivation of a stem-final long vowel

| | | /takabaː/ | NONFIN | WSP | MAX-Vː | BD-DEP-V | FIN-C | MAX-$\mu$ |
|---|---|---|---|---|---|---|---|---|
| a. | | tákab | | | *! | | | ** |
| b. | ☞ | tákaba | | | | | * | * |
| c. | | takabáː | *! | | | | | |
| d. | | tákabaː | | *! | | | | |

Tableau (38) shows that an underlyingly short word-final vowel is deleted, in order to satisfy FIN-C, the markedness constraint that requires words to end with a consonant. Tableau (39) shows that an underlyingly long vowel in the same position is shortened. If the long vowel had surfaced as long, it would have caused a violation of the stress markedness constraints NONFINALITY and WSP. But because MAX-Vː outranks FIN-C, the optimal solution is to shorten the vowel rather than deleting it.

Consider now the same vowel-final stem before a suffix. Tableau (40) shows the derivation where the stem-final vowel is underlyingly short. In this case, BD-DEP-V kicks in and rules out candidate (40b) because the base [tákab] (the output of (38)) does not end with a vowel.

(40) McCarthy's (2005) derivation of a suffixed stem with a short final vowel

| | | | NONFIN | WSP | MAX-Vː | BD-DEP-V | FIN-C | MAX-$\mu$ |
|---|---|---|---|---|---|---|---|---|
| | | /takaba-haː/ | | | | | | |
| | *Base*: | [tákab] | | | | | | |
| a. | ☞ | takábha | | | | | * | * |
| b. | | takábaha | | | | *! | * | |

On the other hand, if the stem-final vowel is underlyingly long, as in tableau (41), BD-DEP-V is irrelevant because the base [tákaba] (the output of (39)) ends with a vowel. The stem-final vowel remains long and receives stress on the surface.

(41) McCarthy's (2005) derivation of a suffixed stem with a long final vowel

| | | | NONFIN | WSP | MAX-Vː | BD-DEP-V | FIN-C | MAX-$\mu$ |
|---|---|---|---|---|---|---|---|---|
| | | /takabaː-haː/ | | | | | | |
| | *Base*: | [tákaba] | | | | | | |
| a. | ☞ | takabáː-ha | | | | | * | * |
| b. | | takaba-háː | *! | | | | * | * |
| c. | | takabáː-haː | | *! | | | * | |

McCarthy (2005) suggested that his analysis should also work for longer morpheme configurations, but this is not always the case, as can be seen by considering the NZA example [nsi-na-háː-ʃ] we have used so far. We show this in tableau (42), which uses McCarthy's constraints and the UR /nsiː-naː-haː-ʃ/. The incorrect candidate (42a) is favored by McCarthy's constraints as it has strictly fewer violations.

(42) McCarthy's (2005) analysis fails on words with two pre-boundary vowels

| | | | NONFIN | WSP | MAX-Vː | BD-DEP-V | FIN-C | MAX-$\mu$ |
|---|---|---|---|---|---|---|---|---|
| | | /nsiː-naː-haː-ʃ/ | | | | | | |
| | *Base*: | [nsiná:ha] | | | | | | |
| a. | ☹ | nsi-na-háː-ʃ | *! | | | | | * |
| b. | ☞ | nsi-náː-ha-ʃ | | | | | | * |

To summarize this section, McCarthy (2005) has shown that B-D correspondence succeeds in generating forms that consist of a stem and one suffix. However, as tableau (42) illustrates, McCarthy's analysis fails when more than one suffix attaches to the stem. Note that this failure is not something that can be fixed by re-ranking the constraints in (42). The correct output in (42a) is harmonically bounded by the candidate in (42b) because the violation profile of (42a) is a proper superset of that of (42b).

# 5 Stratal OT analysis

This section constitutes the final part of our theoretical argument. In what follows, we show that the success of the rule-based analysis in section 2.2 does indeed stem from the architectural properties presented in (5), and repeated here in (43). We do so by providing an alternative analysis within Stratal OT (Kiparsky, 2000, 2015; Bermúdez-Otero, 2011) – a theoretical framework that is fundamentally different than rule-based phonology, but shares the properties in (43). These properties are exactly what distinguishes Stratal OT from the parallel model of OT discussed previously, which fails to account for the NZA pattern.

(43) Architectural properties required to generate the NZA pattern

a. A serial account of cyclicity.

b. The ability to erase morphological boundaries between cycles.

There are multiple versions of Stratal OT that differ from each other with respect to properties such as the number of strata, which strata are cyclic, and so on. But the properties in (44) are shared by all versions of the theory we know of.

(44) Properties of Stratal OT

a. The output of constraint evaluation in one stratum serves as the input to constraint evaluation in a following stratum.

b. A ranking of $C_1 \gg C_2$ between two constraints in one stratum may be replaced with the opposite ranking $C_2 \gg C_1$ in another stratum.

In fact, the analysis we are about to sketch does not rely on the property in (44b), as we will not need to change the constraint rankings between cycles. The crucial property needed for the analysis to work is the one in (44a), which enables the repeated application of the same process in cycles defined by the morphological structure. As will be shown immediately, this property, combined with the ability to erase internal morpheme boundaries between cycles, will be enough to generate the correct NZA surface forms.

The analysis involves the stress-related constraints from the previous sections, as well as two markedness constraints that were presented in section 3 and are given here again for convenience:

(45)

- $^*\breve{V}+$ - assign * for every short vowel that is immediately followed by a morpheme boundary.
- $^*V{:}C\acute{V}{:}$ - assign * for every long vowel in an open syllable that precedes a stressed long vowel.

In Parallel OT, these independently motivated constraints are not useful because, as long as all morpheme boundaries are persistent throughout the derivation, all three conceivable candidates are equally harmonic with respect to the constraints in (45). As shown earlier in (27), [nsi-na-háː-ʃ], [nsi-náː-ha-ʃ], and [nsíː-na-ha-ʃ] all violate $^*\breve{V}+$ twice, and satisfy $^*V{:}C\acute{V}{:}$. This is going to change once we move to a serial architecture in which only one morphological boundary is available in each derivational step.

Consider the tableaux in (46), which illustrate the derivation of [nsinaháːʃ] in three cycles.

(46) A cyclic derivation of [nsinaháːʃ] 'we didn't forget her' in Stratal OT

a. Cycle $\varphi_1$

| $\varphi_1$ /nsiː-naː/ | $^*\breve{V}+$ | WSP Super-heavy | NON FINALITY | WSP heavy | EXNON FINALITY | $^*V{:}C\acute{V}{:}$ |
|---|---|---|---|---|---|---|
| a. ☞ nsíː-na | | | | | * | |
| b. nsí-na | *! | | | | * | |

b. Cycle $\varphi_2$

| $\varphi_2$ /nsí:na-ha:/ | $^*\breve{V}+$ | WSP Super-heavy | NON FINALITY | WSP heavy | EXNON FINALITY | $^*V{:}C\acute{V}{:}$ |
|---|---|---|---|---|---|---|
| a. ☞ nsiná:-ha | | | | | * | |
| b. nsí:na-ha | *! | | | | | |
| c. nsi:ná:-ha | | | | | * | *! |

c. Cycle $\varphi_3$

| $\varphi_3$ /nsiná:ha-ʃ/ | $^*\breve{V}+$ | WSP Super-heavy | NON FINALITY | WSP heavy | EXNON FINALITY | $^*V{:}C\acute{V}{:}$ |
|---|---|---|---|---|---|---|
| a. ☞ nsinahá:-ʃ | | | * | | | |
| b. nsiná:ha-ʃ | *! | | | | * | |
| c. nsina:há:-ʃ | | | * | | | *! |

As shown in (46a), in the derivational cycle of the stem and the subject marker, [nsí:-na], $^*\breve{V}+$ determines that the stem-final vowel should surface as long before the morphological boundary. The output of the first cycle serves as the input to the evaluation in (46b), alongside the object clitic that enters the derivation in this stage. Obeying the Bracket Erasure convention, the nested morpheme boundary is deleted at the end of the first cycle, and the only morphological boundary present in the second cycle is the boundary before the object clitic. Therefore, unlike in Parallel OT, here $^*\breve{V}+$ can only target the rightmost vowel before an internal morpheme boundary. As can be seen in (46b), because $^*\breve{V}+$ outranks the stress-related constraints, it is better to lengthen the penultimate vowel and assign stress to it rather than assign the default antepenultimate stress. If the embedded morpheme boundary could have triggered violations of $^*\breve{V}+$, it would have been violated by the winning candidate as well and there would have been no way to favor this candidate over the other candidates (as in the derivation in 27). In the second cycle, $^*V{:}C\acute{V}{:}$ also becomes active and motivates the shortening of the pretonic vowel. The output of the second cycle serves as the input to the third, where the negation suffix is added. As shown in (46c), the output of the third cycle is determined similarly to the second cycle, albeit with ultimate stress.[14]

As just seen, the NZA pattern does not pose a challenge to Stratal OT. Consequently, the logical conclusion is that the challenge posed by the pattern to Parallel OT has nothing to do with the optimization process, the set of constraints, or any other property shared by Stratal OT and Parallel OT. The relevant theoretical points of divergence with respect to the NZA pattern are the serial account of cyclicity and the ability to erase morphological boundaries by the end of cycles – properties that Stratal OT has (as well as rule-based phonology), but Parallel OT, with or without B-D correspondence, does not. We therefore argue that phonological computation is cyclic and does not serve the purpose of enforcing similarity between related words.

[14]For simplicity, we only show derivations for URs containing long vowels. However, as long as the relevant faithfulness constraints are ranked below the six markedness constraints used here, the analysis should work given all possible underlying length specifications.

# 6 The pattern is fully general

Our goal in this section is to show that the interaction of stress and vowel length in NZA as described in subsection 2.1 is fully general and goes beyond the examples presented before. Consider the word [btins-u-háː-ʃ] '*you are not forgetting her*', which exemplifies the generalization in (12) as a vowel preceding a morpheme boundary is long only where it would receive stress and otherwise short. The relevant property of this word is that it includes a sequence of three morphemes $M_1$, $M_2$, $M_3$ such that $M_1$ and $M_2$ are vowel-final and the entire sequence occurs word-finally. The existence of $M_3$ and the fact that the sequence occurs word-finally causes stress to fall on the final vowel of $M_2$. The occurrence of the vowel-final $M_1$ in the sequence results in a configuration where some vowel occurs before a morpheme boundary in a position where it does not receive stress. In what follows, we describe all the different morpheme sequences in NZA verbs we know of that have this property.

The general structure of the NZA verb is given in (47). The portion that is relevant for our purposes is the portion that includes the stem and the suffixes. Suffixes can include subject agreement markers, object clitics, and the negation suffix, in this order.

(47) Structure of the NZA verb
(Prefixes)-STEM-(SUBJECT.AGR)-(OBJECT.CLITIC)-(NEG)

Morphemes that can be vowel-final are the stem, the perfective subject agreement markers -ti '2.sg.f', -na '1.pl', -tu '2.pl', and -u '3.pl', the imperfective subject agreement markers -i '2.sg.f' and -u '2.pl' or '3.pl', and the direct object clitics -ni '1.sg', -ki '2.sg.f', -ka '2.sg.m', -ha '3.sg.f', -na '1.pl', and -ku '2.pl'. Together we get four types of morpheme combinations with the relevant properties, which are listed in the following table. In this table and below we keep using the stem of the verb 'forget' for illustration, and the negation suffix is -ʃ.

(48) Morpheme sequences with two adjacent vowel-final morphemes preceded by a word-final suffix.

| Type | $M_1$ (vowel-final) | $M_2$ (vowel-final) | $M_3$ | Example | Alternative |
|---|---|---|---|---|---|
| I | SUBJECT.AGR | OBJECT.CLITIC | NEG | btins-u-háː-ʃ | *btins-úː-ha-ʃ |
| II | STEM | OBJECT.CLITIC | NEG | btinsa-níː-ʃ | *btins-áː-ni-ʃ |
| III | STEM | SUBJECT.AGR | OBJECT.CLITIC | nsi-túː-ha | *ns-íː-tu-ha |
| IV | STEM | SUBJECT.AGR | NEG | nsi-náː-ʃ | *ns-íː-na-ʃ |

The full range of examples is given below – a total of 74 different morpheme combinations – divided by type.

In Type I examples, which we present in two parts below for reasons of formatting, the stem is followed by a vowel-final subject agreement marker, then a vowel-final object clitic, then the negation marker (the stem is also vowel final in some of these examples, but here we focus on the suffixes and the relationship between them and disregard the stem). The examples below are all the possible combinations of this type (with the verb 'forget'). The running example in the paper, [nsi-na-háː-ʃ] '*we did not forget her*', is of this type, so each of these examples share the attributes that will have theoretical implications in section 3.

Here we should note that additional processes might be involved in the derivation of the surface stem of verbs like '*forget*', depending on what the UR of the stem is, a question that there is no agreement on in the literature on Arabic. Vowel-final stems seem to come from tri-consonantal roots with a final glide (e.g., the root of 'forget' is n.s.j) which surfaces in some environments (e.g., [nís.ju] '*they forgot*') but not in others (as in most examples in this paper). If the UR of the stem includes a final glide, then a process that deletes or alters the final glide would have to take place in addition to lengthening before the object clitic (e.g., /nisij-na/ → [nsiːna] '*we forgot*'). As far as we can tell, the possibility of an underlying glide and a process that deletes or alters it does not affect the main argument of the paper. The same goes for the deletion of the first vowel in /nisij-na/ → [nsiːna].

(49) Type I examples – part 1 (total: 18/36 examples in part 1)

a. ma nsi-ti-níː-ʃ 'You.f didn't forget me'
b. ma nsi-ti-kíː-ʃ 'You.f didn't forget you.f'
c. ma nsi-ti-káː-ʃ 'You.f didn't forget you.m'
d. ma nsi-ti-háː-ʃ 'You.f didn't forget her'
e. ma nsi-ti-náː-ʃ 'You.f didn't forget us'
f. ma nsi-ti-kúː-ʃ 'You.f didn't forget you.pl'
g. ma nsi-na-níː-ʃ 'We didn't forget me'
h. ma nsi-na-kíː-ʃ 'We didn't forget you.f'
i. ma nsi-na-káː-ʃ 'We didn't forget you.m'
j. ma nsi-na-háː-ʃ 'We didn't forget her'
k. ma nsi-na-náː-ʃ 'We didn't forget us'
l. ma nsi-na-kúː-ʃ 'We didn't forget you.pl'
m. ma nsi-tu-níː-ʃ 'You.pl didn't forget me'
n. ma nsi-tu-kíː-ʃ 'You.pl didn't forget you.f'
o. ma nsi-tu-káː-ʃ 'You.pl didn't forget you.m'
p. ma nsi-tu-háː-ʃ 'You.pl didn't forget her'
q. ma nsi-tu-náː-ʃ 'You.pl didn't forget us'
r. ma nsi-tu-kúː-ʃ 'You.pl didn't forget you.pl'

(50) Type I examples: part 2 (total: 18/36 examples in part 2)

a. ma nisj-u-níː-ʃ 'They.pl didn't forget me'
b. ma nisj-u-kíː-ʃ 'They.pl didn't forget you.f'
c. ma nisj-u-káː-ʃ 'They.pl didn't forget you.m'
d. ma nisj-u-háː-ʃ 'They.pl didn't forget you.m'
e. ma nisj-u-náː-ʃ 'They.pl didn't forget us'
f. ma nisj-u-kúː-ʃ 'They.pl didn't forget you.pl'
g. btins-i-níː-ʃ 'You.f are not forgetting me'
h. btins-i-kíː-ʃ 'You.f are not forgetting you.f'
i. btins-i-káː-ʃ 'You.f are not forgetting you.m'

j. btins-i-háː-ʃ 'You.f are not forgetting her'
k. btins-i-náː-ʃ 'You.f are not forgetting us'
l. btins-i-kúː-ʃ 'You.f are not forgetting you.pl'
m. btins-u-níː-ʃ 'You.pl are not forgetting me'
n. btins-u-kíː-ʃ 'You.pl are not forgetting you.f'
o. btins-u-káː-ʃ 'You.pl are not forgetting you.m'
p. btins-u-háː-ʃ 'You.pl are not forgetting her'
q. btins-u-náː-ʃ 'You.pl are not forgetting us'
r. btins-u-kúː-ʃ 'You.pl are not forgetting you.pl'

In Type II examples, the sequence of relevant morphemes consists of a vowel-final stem, followed by a vowel-final object clitic, which is in turn followed by the negation marker. Vowel lengthening applies to stem-final vowels before an object clitic (e.g., [nisíː-ni] 'he forgot me') so these examples show the same pattern as before: the only long vowel is the rightmost vowel that precedes a morpheme boundary (in this case, the vowel of the object clitic).

(51) Type II examples (total: 12 examples)
a. ma nisi-níː-ʃ 'he didn't forget me'
b. ma nisi-kíː-ʃ 'he didn't forget you.f'
c. ma nisi-káː-ʃ 'he didn't forget you.m'
d. ma nisi-háː-ʃ 'he didn't forget her'
e. ma nisi-náː-ʃ 'he didn't forget us'
f. ma nisi-kúː-ʃ 'he didn't forget you.pl'
g. binsa-níː-ʃ 'he is not forgetting me'
h. binsa-kíː-ʃ 'he is not forgetting you.f'
i. binsa-káː-ʃ 'he is not forgetting you.m'
j. binsa-háː-ʃ 'he is not forgetting her'
k. binsa-náː-ʃ 'he is not forgetting us'
l. binsa-kúː-ʃ 'he is not forgetting you.pl'

In Type III examples, the sequence of relevant morphemes consists of a vowel-final stem, followed by a vowel-final subject agreement marker, which is in turn followed by an object clitic. Type III has the same configuration that was shown in (11b) but the examples below show that the pattern is exceptionless for all vowel-final subject markers.

(52) Type III examples (total: 23 examples)
a. nsi-tíː-ni 'you.f forgot me'
b. nsi-tíː-ki 'you.f forgot you.f'
c. nsi-tíː-ha 'you.f forgot her'
d. nsi-tíː-(h) 'you.f forgot him'
e. nsi-tíː-na 'you.f forgot us'

f. nsi-tíː-ku 'you.f forgot you.pl'
g. nsi-tíː-hen 'you.f forgot them.pl'
h. nsi-náː-ni 'we forgot me'
i. nsi-náː-ki 'we forgot you.f'
j. nsi-náː-k 'we forgot you.m'
k. nsi-náː-ha 'we forgot her'
l. nsi-náː-(h) 'we forgot him'
m. nsi-náː-na 'we forgot us'
n. nsi-náː-ku 'we forgot you.pl'
o. nsi-náː-hen 'we forgot them'
p. nsi-túː-ni 'you.pl forgot me'
q. nsi-túː-ki 'you.pl forgot you.f'
r. nsi-túː-k 'you.pl forgot you.m'
s. nsi-túː-ha 'you.pl forgot her'
t. nsi-túː-(h) 'you.pl forgot him'
u. nsi-túː-na 'you.pl forgot us'
v. nsi-túː-ku 'you.pl forgot you.pl'
w. nsi-túː-hen 'you.pl forgot them'

Finally, in Type IV examples, a vowel-final stem is followed by a subject marker and then negation.

(53) Type IV examples (total: 3 examples)
a. ma nsi-tíː-ʃ 'you.f didn't forget'
b. ma nsi-náː-ʃ 'we didn't forget'
c. ma nsi-túː-ʃ 'you.pl didn't forget'

In sum, the generalization regarding stress and vowel length in (12) holds as expected in all relevant morpheme combinations that we could think of, without exception. This suggests that the pattern is fully general and is not limited to particular morphemes or morpheme combinations.

# 7 Conclusion

Cyclic patterns in phonology have received two fundamentally different theoretical explanations that have been difficult to distinguish empirically: the cycle and B-D Correspondence. The NZA pattern provides evidence for a cyclic grammatical component because the application of NZA processes decreases the similarity between related surface forms. As mentioned in the introduction, some previous work has made the opposite claim, which is that B-D Correspondence is required in cases where the cycle is insufficient. If this claim is correct, it would mean that similarity between surface forms is not just an epiphenomenon of cyclic computation. And given the argument of the current paper, the correctness of that claim would lead to the unfortunate conclusion that both the cycle and B-D Correspondence – two mechanisms designed to deal with the same phenomenon – should coexist in the grammar. We leave it for future work to determine whether this conclusion is necessary.

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